

Deep Learning and Applications

Course 6

Graphs - Diffusion - Fairness

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Semester 1

Today's content

- ▶ Short Introduction to Graph Neural Networks
- ▶ Introduction to Diffusion models
- ▶ Introduction to Ethics and fairness

Credits: The Fidle courses on Graph Neural Networks and Diffusion models (<https://fidle.cnrs.fr>).

Lilian Weng blog on diffusion models,

<https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>,

Charlotte Laclau's slides on Ethics and fairness

Graph Neural Networks

A very short introduction to Graph Neural Networks

We use the slides of the Fidle course on Graph Neural Networks (09),
<https://fidle.cnrs.fr>

Graphs store information: Labels

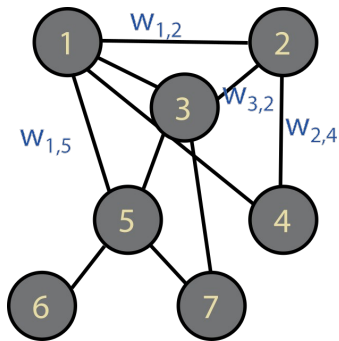
- Graphs can store information (features) on **nodes**, **edges** and **globally**

	Globally	Nodes	Edges
Social Network	Group of interest, ...	Name, Age, Job, ...	Is friend, follows, family, ...
Molecule	Is a drug, Energy, ...	Atomic number, ...	Bond order, ...
Citations	Field, ...	Article, ...	Was cited, ...
Particle physics	Experiment	Particle	Decayed to, ...
Motion capture	Character	Joints	Is connected to, ...
Natural language	Paragraph, ...	Group of words, ...	Refers to, ...

- It can be a number, a concept, ...

Node proximity and centrality

Measure of the structure of a graph



Node proximity

- 1st order: $w_{i,j}$ between node i and j
- 2nd order: similarity of neighborhood structure
- Higher orders possible

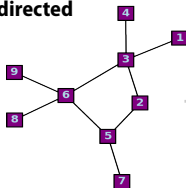
Node centrality

- Measure how many paths goes through the node

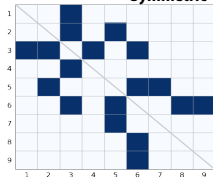
Graph representation

Adjacency matrix $W_{(i,j)} = \begin{cases} w_{i,j} & \text{if there is an edge} \\ 0 & \text{if not} \end{cases}$

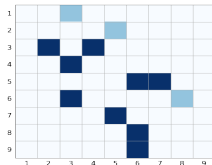
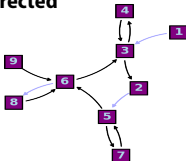
Undirected



Symmetric

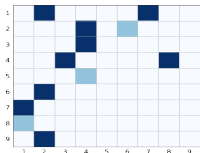
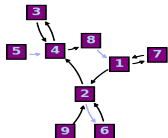


Directed



Graph representation

Adjacency list



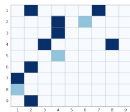
Nodes: [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0]

Edges: [0.4, 0.4, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 0.4, 1.0, 1.0, 1.0]

Adjacency list: [[5, 4],
[8, 1],
[4, 8], [4, 3],
[3, 4],
[1, 7], [1, 2],
[2, 4], [2, 6],
[7, 1],
[6, 2],
[9, 2]]

Global: [1.0, 1.0]

Graph representation



- Scale $V^2 \rightarrow$ lot of space
- Sparse
- $N!$ permutations to represent the same graph
- Easy to find an edge

Adjacency list: [[5, 4],
[8, 1],
[4, 8], [4, 3],
[3, 4],
[1, 7], [1, 2],
[2, 4], [2, 6],
[7, 1],
[6, 2],
[9, 2]]

- Scale $E \rightarrow$ less space
- Might be difficult to find an edge

V = number of nodes/vertices
 E = number of edges

<https://www.geeksforgeeks.org/comparison-between-adjacency-list-and-adjacency-matrix-representation-of-graph/>

Useful Matrices

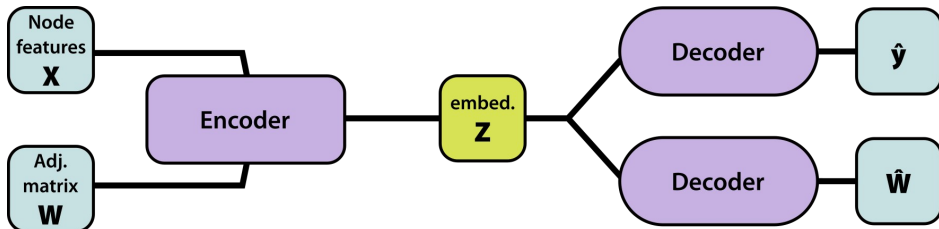
Adjacency	W	Weight of edges
Degree	D	Diagonal matrix with number of edges for each node
Laplacian	L	D - W
Node Features	X	Information stored

Learning on Graphs

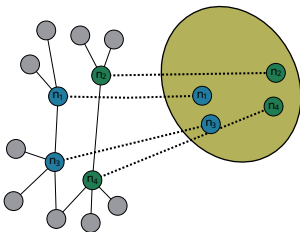
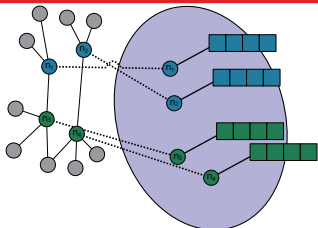


Graph embedding

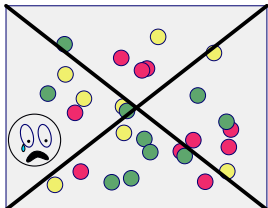
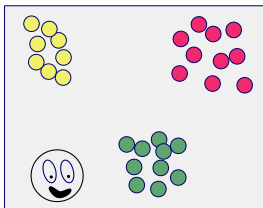
- We need to find a representation of the graph that is processable



Graph embedding



- Features stored in nodes/edges/graphs are not easily processed.
- We transform the features into a vector in the latent space (**Dimension is a hyperparameter**).
- The embedding has to be suited for the task → **Learnable**.

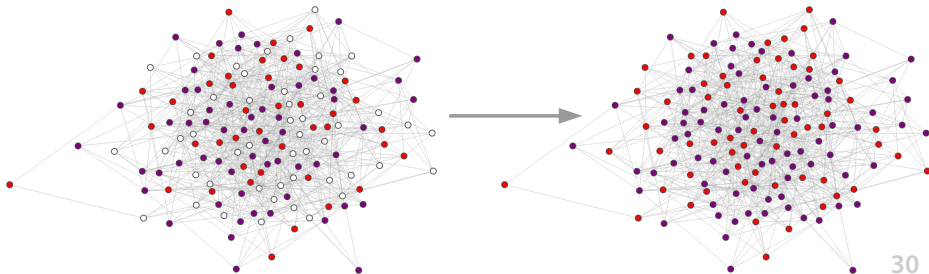


Transductive learning

The model has access to the complete graph

It is not possible to add new nodes

Node labeling

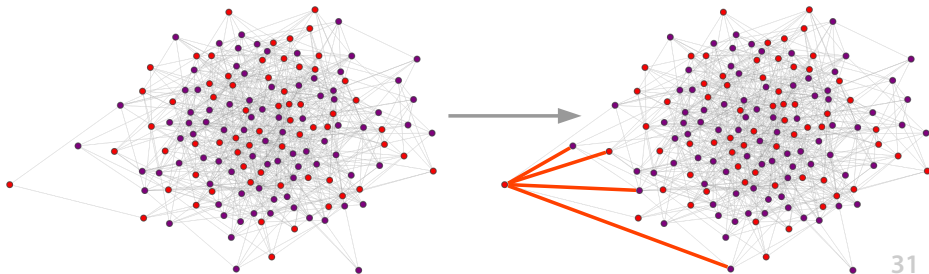


Transductive learning

The model has access to the complete graph

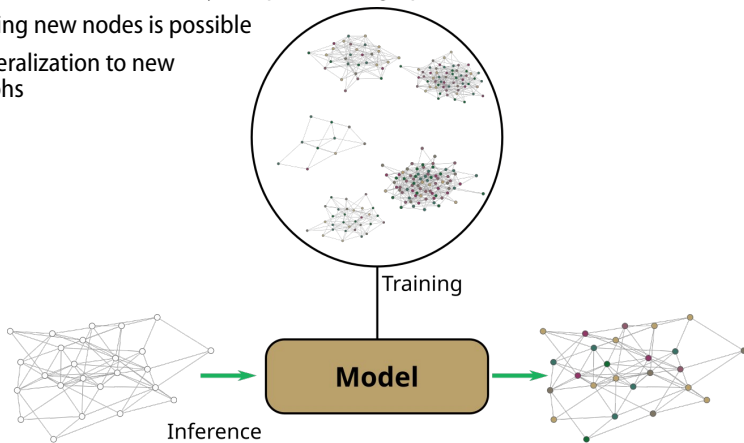
It is not possible to add new nodes

Find new edges



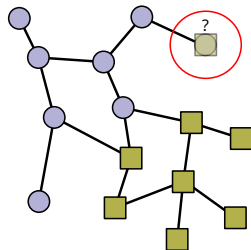
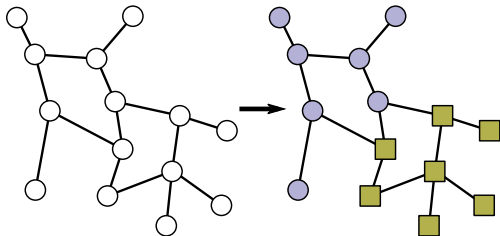
Inductive learning

- The model has access only to a part of the graph (train set)
- Adding new nodes is possible
- Generalization to new graphs



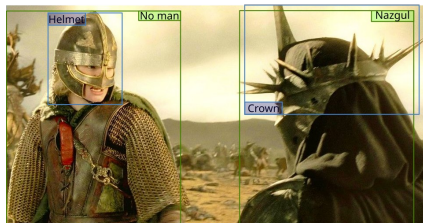
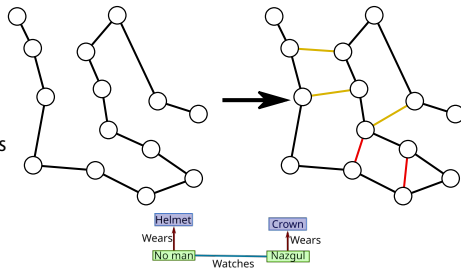
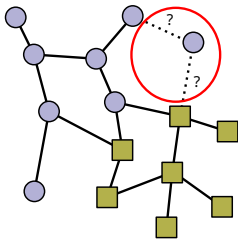
Tasks on nodes

- Labeling nodes in a graph
(clustering)
 - Find topic of a research paper (CORA, etc)
 - Find bots in a social network
 - ...
- Labeling new nodes
- Perform regression



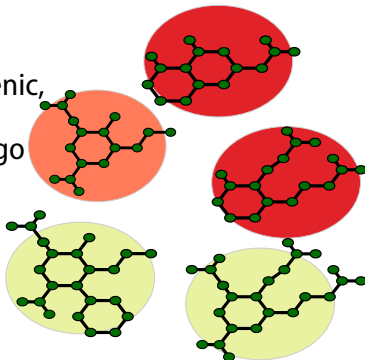
Tasks on edges

- Find relationships
 - Contact map of aminoacids (Alphafold)
 - Contact suggestion (social network)
 - ETA for directions (regression)
 - Relationships between segments in pictures
 - ...



Tasks on graphs

- Predict properties of graphs
 - Chemical properties (solubility, carcinogenic, possible drug)
 - Classification of the research field in an ego network
 - ...



A few examples

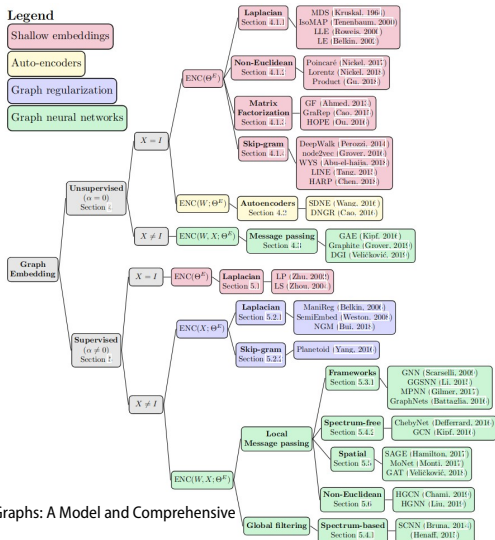


Taxonomy of methods



Legend

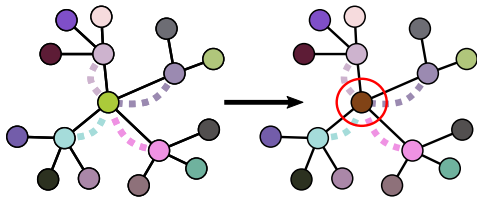
- Shallow embeddings
- Auto-encoders
- Graph regularization
- Graph neural networks



I. Chami, S. Abu-El-Haija, and B. Perozzi, "Machine Learning on Graphs: A Model and Comprehensive Taxonomy".

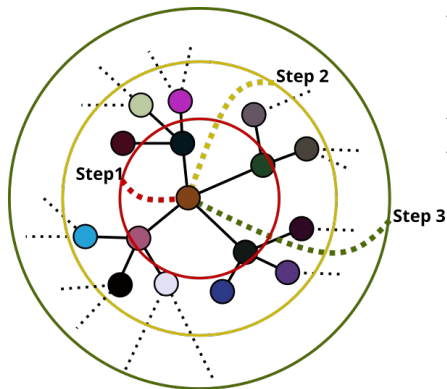
Graph convolution

- Just like for images we can learn from neighborhood with a convolution operator.



- A bit more complex since the number of neighbors is unlikely to be constant.
- We want the operator to be permutation invariant.

Graph convolution



- Several steps are needed to retrieve information for distant nodes.
- For large graphs → a **cutoff**
- It is possible to use a **virtual node** connected to all other nodes. But in practice this becomes quickly intractable.

Example: Graph Convolution Networks (GNN)

Principle - message passing

- ▶ Aggregating over immediate neighbour features
- ▶ Combining with the node's own feature
- ▶ Let $h_v^{(i)}$ the embedding of node v at layer i

Architecture, for all node v

- ▶ Layer 0: $h_v^{(0)} = x_v$ - original features of node v
- ▶ Layer k : $h_v^{(k)} = f^{(k)} \left(W^{(k)} \cdot \frac{\sum_{u \in \mathcal{N}(v)} h_u^{(k-1)}}{|\mathcal{N}(v)|} + B^{(k)} \cdot h_v^{(k-1)} \right)$
 - ▶ $f^{(k)}$: activation function (potentially learnable), $W^{(k)}, B^{(k)}$: parameters to be learned
 - ▶ $h_u^{(k-1)}$ embeddings of the **neighbors of** v in the previous layer
 - ▶ $h_v^{(k-1)}$ previous embedding of v

Message passing

- Predictions can be made at each node from the final layer:
 $\hat{y}_v = \text{PREDICT}(h_v^{(K)})$, where *PREDICT* is another network (e.g. fully connected) to be learned.

- Other strategies:

- Graph Attention Networks GAT:

$$h_v^{(k)} = f^{(k)} \left(W^{(k)} \cdot \left[\sum_{u \in \mathcal{N}(v)} \alpha_{vu}^{(k-1)} h_u^{(k-1)} + \alpha_{vv}^{(k-1)} h_v^{(k-1)} \right] \right)$$
$$\alpha_{vu}^{(k)} = \frac{A^{(k)}(h_v^{(k)}, h_u^{(k-1)})}{\sum_{u \in \mathcal{N}(v)} A^{(k)}(h_v^{(k)}, h_u^{(k-1)})} \text{ with } A^{(k)} \text{ an attention mechanism}$$

- Graph Sample and Aggregate GraphSAGE:

$$h_v^{(k)} = f^{(k)} \left(W^{(k)} \cdot \left[\text{AGG}_{u \in \mathcal{N}(v)}(\{h_u^{(k-1)}\}), h_v^{(k-1)} \right] \right)$$

with aggregation of nodes

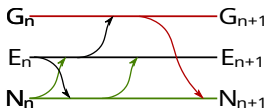
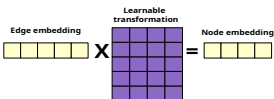
- Graph Isomorphism Network GIN:

$$h_v^{(k)} = f^{(k)} \left(\sum_{u \in \mathcal{N}(v)} h_u^{(k-1)} + (1 + \epsilon^{(k)}) \cdot h_v^{(k-1)} \right).$$

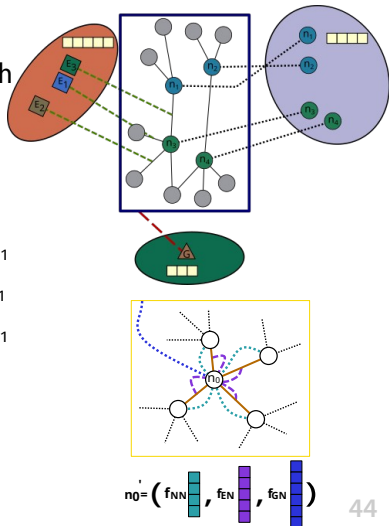
We can imagine to add other strategies: pooling (selecting one node among the neighbors), merging nodes to develop a hierarchical architecture...

Message passing

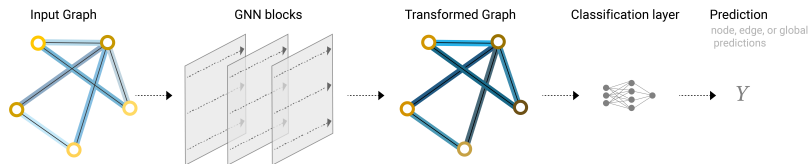
- We have embeddings for each part of the graph (possibly different vector sizes).
- Each part can learn from the others via a transformation.



- Information is aggregated to form a message that the node/edge will send to others.



Final Illustration



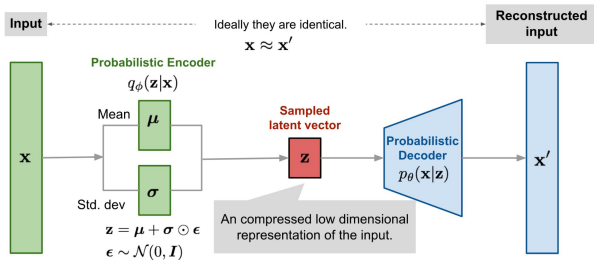
From the blog of Sanchez-Lengeling, et al., "A Gentle Introduction to Graph Neural Networks", <https://distill.pub/2021/gnn-intro/>

Diffusion Probabilistic Models (DPM)

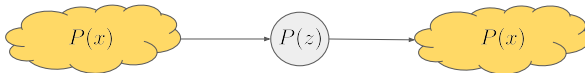
Denoising Diffusion Probabilistic Models (DDPM)

We use the slides of the Fidle course on Diffusion models (14),
<https://fidle.cnrs.fr>

Rappel - VAE - 1



Source <https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>



Rappel - VAE - 2

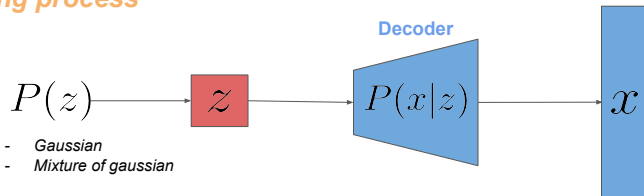
Training process

metric

VAE COST FUNCTION

$$\log p_{\theta}(\mathbf{x}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x})||p_{\theta}(\mathbf{z}|\mathbf{x})) = \mathbb{E}_{\mathbf{z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})} \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x})||p_{\theta}(\mathbf{z}))$$

Sampling process



Rappel - GAN - 1

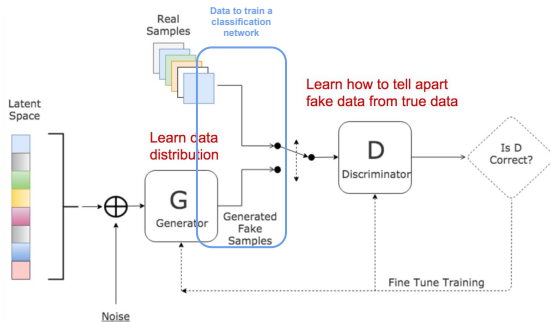
Adding supervision concept in an unsupervised task !

2 networks in opposition :

- Generator
- Discriminator

Ideal solution :

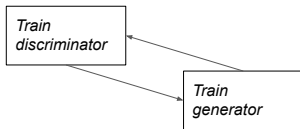
- Generator $\sim P(x|z)$
- Discriminator = $\frac{1}{2}$



Source
<https://www.kdnuggets.com/2017/01/generative-adversarial-networks-hot-topic-machine-learning.html>

Rappel - GAN - 2

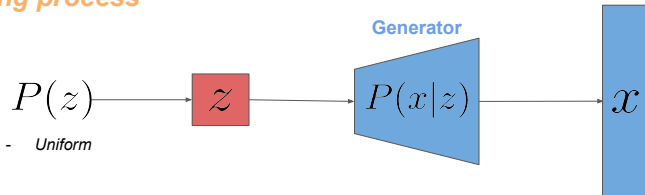
Training process



GAN COST FUNCTION

$$\begin{aligned}\min_G \max_D L(D, G) &= \mathbb{E}_{x \sim p_r(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))] \\ &= \mathbb{E}_{x \sim p_r(x)} [\log D(x)] + \mathbb{E}_{x \sim p_g(x)} [\log(1 - D(x))]\end{aligned}$$

Sampling process



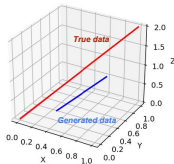
Rappel - GAN - 3

GAN convergence problems !

No convergence due to the nature of the problem (MinMax)

?

Vanishing gradient due to discriminant being too perfect, generator can't train anymore



Source

<https://lilianweng.github.io/posts/2017-08-20-gan/>

Mode collapse due to generator learning only some good examples instead of the whole data distribution

Generated Data

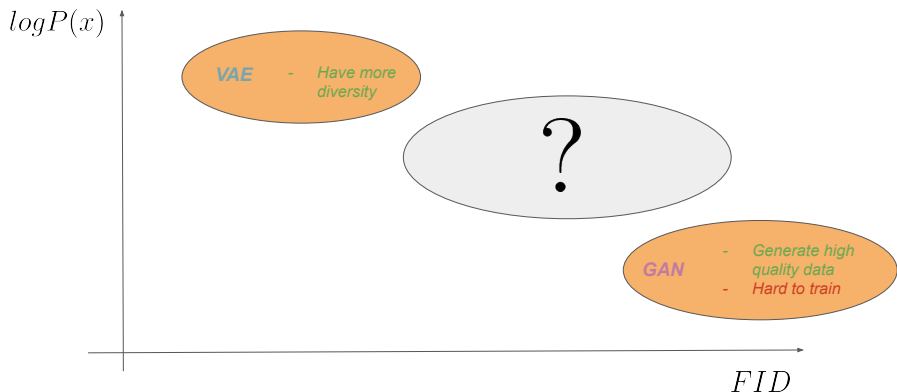


True Data

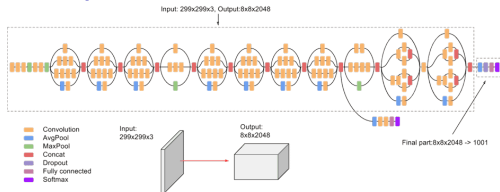


VAE vs GAN

How to compare generative models ?



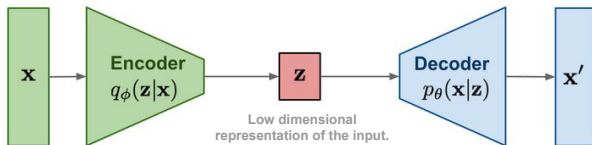
FID: Fréchet-Inception Distance



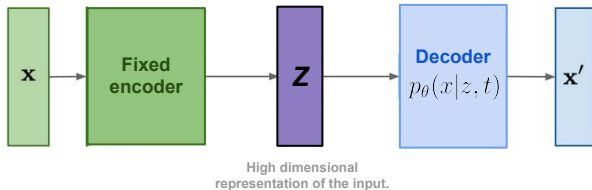
- ▶ We consider 2 sets of images (S_1 - e.g. real - and S_2 generated). Each set is modeled as being drawn from multi-dimensional Gaussian distributions $\mathcal{N}(\mu_i, \Sigma_i)$. We compute the 2-Wasserstein distance between Gaussian distributions (closed form).
- ▶ Mean and variance are computed from the deepest layer of the *inception-v3* architecture: compares how often the same high-level features are found within the 2-sets of images.
- ▶ Distance: $d_F(\mathcal{N}(\mu_1, \Sigma_1), \mathcal{N}(\mu_2, \Sigma_2)) = \|\mu_1 - \mu_2\|_2^2 + \text{Tr} \left(\Sigma_1 + \Sigma_2 - 2(\Sigma_1 + \Sigma_2)^{1/2} \right)$
- ▶ Higher distances indicate a poorer generative model. A score of 0 indicates a perfect model.

VAE vs DPM

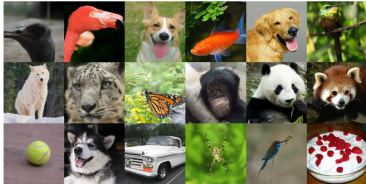
VAE



DPM



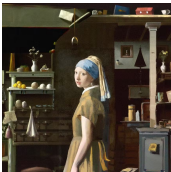
DPM - Landscape



Dhariwal & Nichol, 2021



Source : <https://github.com/bentoml/stable-diffusion-bentoml>



Source : DALL-E 2

DDPM - Principle - 1

After the training the Diffusion Model will generate images from Gaussian noise:

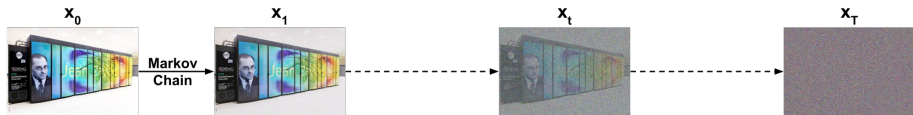


There are three processes that characterize Diffusion Models:

1. Forward Diffusion Process
2. Reverse Diffusion Process
3. Sampling Process

DDPM - Principle - 3

Forward Diffusion Process



This process will add noise to any image gradually

$0 \leq t \leq T$; T is a hyperparameter

DDPM - Principle - 4

Forward Diffusion Process

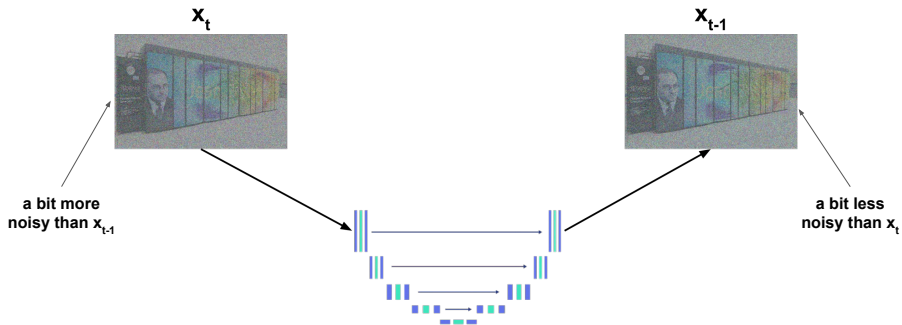


Examples of images at different times t

Here we choose $T=1000$, but it can be different values (it's an hyperparameter)

DDPM - Principle - 5

Reverse Diffusion Process

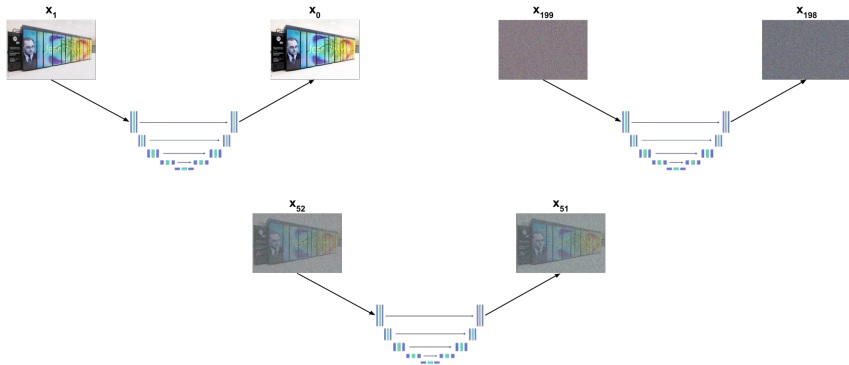


We train a model to predict x_{t-1} from x_t

x_0 is any image from the dataset

DDPM - Principle - 6

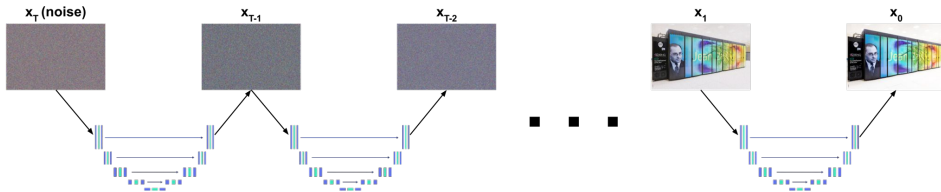
Reverse Diffusion Process



The same model must predict every x_{t-1} from x_t

DDPM - Principle - 7

Sampling Process

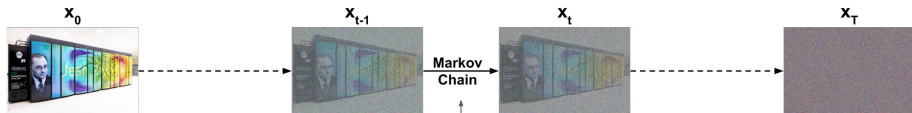


From a random noise we can generate an image

Diffusion process

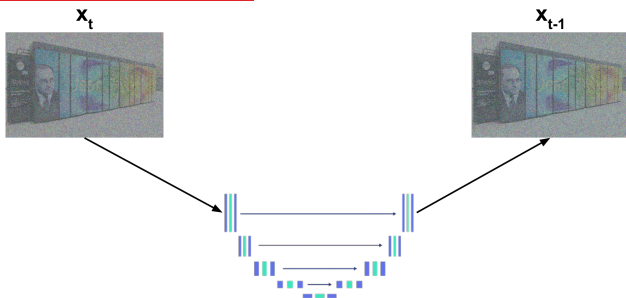
- ▶ We need to model the step of the forward diffusion process $q(x_t|x_{t-1})$:
 - ▶ Markov process depending on a Gaussian transition
 - ▶ $q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t \mathbf{I})$
 - ▶ $\mathcal{N}$ refers to a normal distribution with mean $\sqrt{1 - \beta_t}x_{t-1}$ and variance $\beta_t \mathbf{I}$, $\mathbf{I}$ being the identity matrix.
 - ▶ β_t is called a schedule with values in $[0, 1]$, generally low (e.g. 0.0001 to 0.02) to prevent variance from exploding.
- ▶ Tool - reparametrization trick: $(\mu, \sigma^2) = \mu + \sigma * \epsilon$ with $\epsilon \sim \mathcal{N}(0, 1)$
- ▶ $q(x_{1:T}|x_0) := \prod_{t=1}^T q(x_t|x_{t-1})$
- ▶ Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t := \prod_{s=1}^t \alpha_s$
- ▶ Then $q(x_t|x_{t-1}) = \sqrt{\alpha_t}x_{t-1} + \sqrt{1 - \alpha_t}\epsilon$
- ▶ Finally:
$$q(x_t|x_{t-1}) = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)\mathbf{I})$$
- ▶ $\Rightarrow$ we can simulate noise at any arbitrary step! ($\bar{\alpha}_t$ is known since β_t is defined)

DDPM - Forward Diffusion - 1



$$q(x_t|x_{t-1}) := \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I)$$

DDPM - Reverse Diffusion - 1



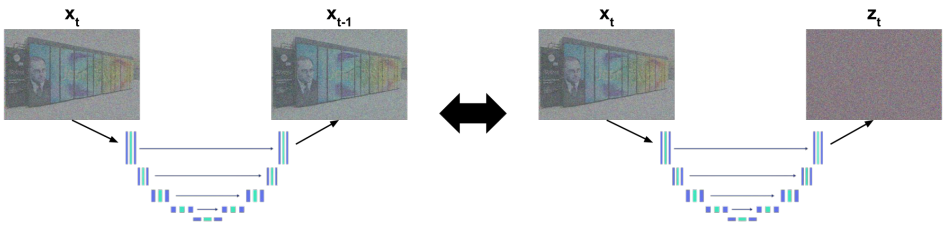
$$q(x_{t-1}|x_t) \approx p_{\theta}(x_{t-1}|x_t)$$

$$p_{\theta}(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_{\theta}(x_t, t), \tilde{\beta}_t I)$$

We predict only the mean, we know the rest.

DDPM - Reverse Diffusion - 2

$$x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} z_t$$



We can predict x_{t-1} by predicting z_t

DDPM - Reverse Diffusion - 3

A little bit of explanation (a tiny bit):

$$q(x_{t-1}|x_t, x_0) = \mathcal{N}(x_{t-1}; \tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t I)$$

$$\tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$$

$$\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}\mathbf{x}_0 + \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{x}_t = \frac{1}{\sqrt{\alpha_t}}(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}}\mathbf{z}_t)$$

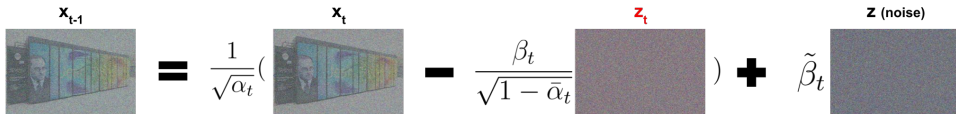


$$q(x_{t-1}|x_t, x_0) \approx p_\theta(x_{t-1}|x_t)$$
$$\frac{1}{\sqrt{\alpha_t}}(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}}\mathbf{z}_t) \approx \frac{1}{\sqrt{\alpha_t}}(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}}\mathbf{z}_\theta(\mathbf{x}_t, \mathbf{t}))$$

DDPM - Reverse Diffusion - 4

$$p(x_{t-1}|x_t, z_t) \rightarrow x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} z_t \right)$$

$\mathbf{x}_{t-1}$ $\mathbf{x}_t$ $\mathbf{z}_t$ $\mathbf{Z}$ (noise)


$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \mathbf{z}_t \right) + \tilde{\beta}_t \mathbf{Z}$$

DDPM - Reverse Diffusion - 5

$$p_{\theta}(x_{t-1}|x_t) \rightarrow x_{t-1} \approx \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} z_{\theta}(x_t, t) \right)$$

Diagram illustrating the reverse diffusion process:

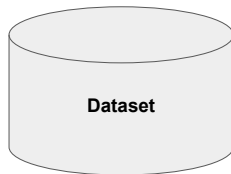
$\mathbf{x}_{t-1} \approx \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \mathbf{z}_{\theta}(\mathbf{x}_t, t) \right) + \tilde{\beta}_t \mathbf{Z} \text{ (noise)}$

The diagram shows the reconstruction of $\mathbf{x}_{t-1}$ from $\mathbf{x}_t$ using the model's output $\mathbf{z}_{\theta}(\mathbf{x}_t, t)$ and added noise $\mathbf{Z}$.

Algorithm 1 Training

- 1: **repeat**
 - 2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
 - 3: $t \sim \text{Uniform}(\{1, \dots, T\})$
 - 4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
 - 5: Take gradient descent step on
$$\nabla_{\boldsymbol{\theta}} \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\boldsymbol{\theta}}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, t) \right\|^2$$
 - 6: **until** converged
-

DDPM - Training - 1



DDPM - Training - 3

**Uniform
distribution**

Between 1 and T



$t = 50$



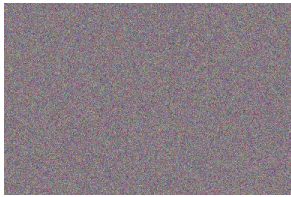
DDPM - Training - 4

**Gaussian
distribution**

Same shape than x_0



$z_t (= \epsilon)$



x_0



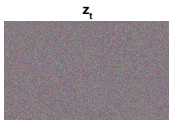
$t = 50$

DDPM - Training - 5

$$\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \mathbf{z}_t$$



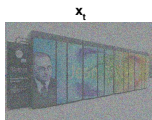
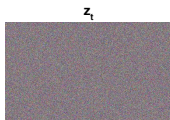

t = 50



DDPM - Training - 5



$t = 50$

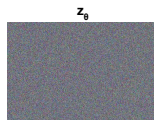
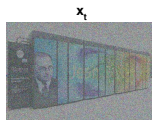
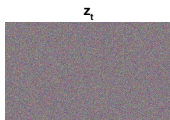


DDPM - Training - 6

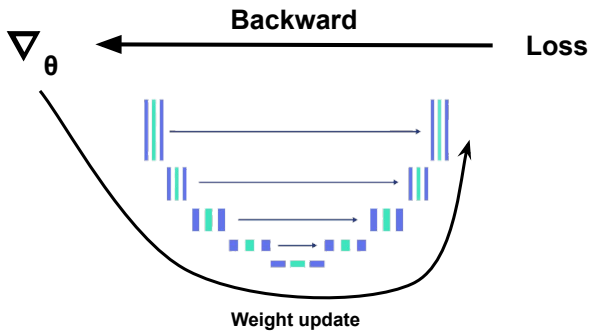
$$\text{Loss} = \left\| z_t - z_\theta \right\|^2$$



$t = 50$



DDPM - Training - 7



and repeat !

It was just 1 iteration.

Algorithm 2 Sampling

```
1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2: for  $t = T, \dots, 1$  do
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$ 
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$ 
5: end for
6: return  $\mathbf{x}_0$ 
```

DDPM - Sampling - 1

**Gaussian
distribution**

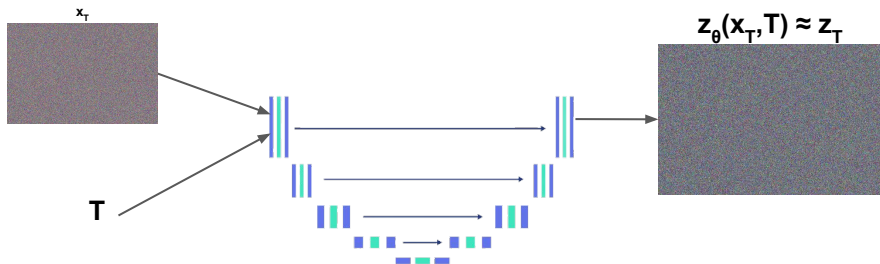
Same shape than training
dataset images



x_T



DDPM - Sampling - 2



Reminder:

$$x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} z_t$$

DDPM - Sampling - 3

**Gaussian
distribution**

Same shape than training
dataset images



z (noise)

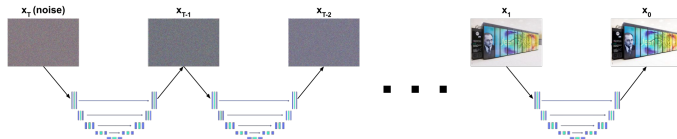


DDPM - Sampling - 4

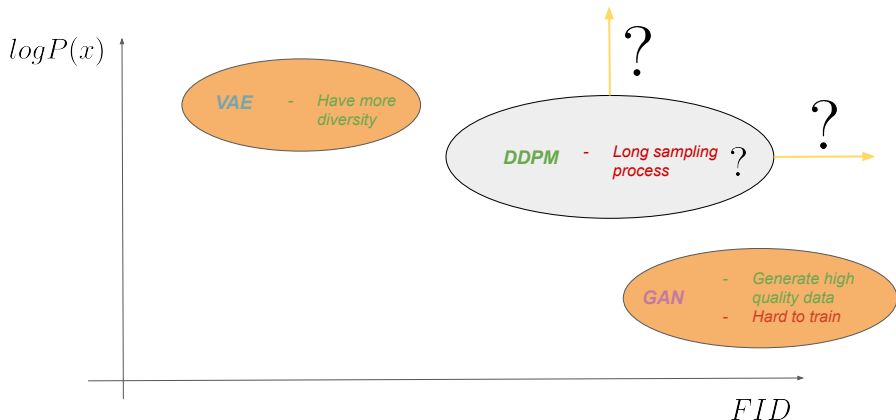
$$\mathbf{x}_{T-1} \approx \frac{1}{\sqrt{\alpha_T}} \left(\mathbf{x}_T - \frac{\beta_T}{\sqrt{1 - \bar{\alpha}_T}} \mathbf{z}_\theta(\mathbf{x}_T, T) \right) + \tilde{\beta}_T \mathbf{z} \text{ (noise)}$$

and repeat !

Don't generate x_T , replace it by x_{T-1} and T by $T-1$... and do it again T time.



DDPM vs VAE vs GAN

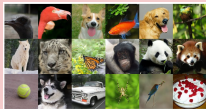


DDPM improvements

Improving the log-likelihood metrics (Improved DDPM, 2021)

Faster sampling
(Denoising Diffusion Implicit Model, 2021 / Latent Diffusion Model, 2021)

Improving image synthesis



Dhariwal & Nichol, 2021

Beta scheduling - Cosine scheduling (IDDPM, 2021)

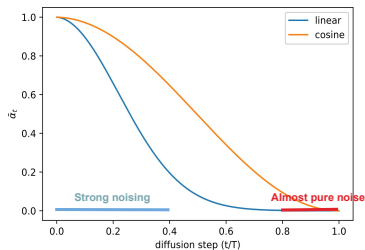
$$(\beta_1, \dots, \beta_T) = (\beta_1 + (t-1) * \frac{\beta_T - \beta_1}{T-1})_{t \in \{1 \dots T\}}$$

$$\alpha_t = 1 - \beta_t \quad \bar{\alpha}_t = \prod_{k=1}^t \alpha_k$$

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I)$$

$$1 > \bar{\alpha}_1 > \dots > \bar{\alpha}_T \approx 0$$

$$q(x_T|x_0) = \mathcal{N}(x_T; 0, I)$$



$$\bar{\alpha}_t = \frac{f(t)}{f(0)}, \quad f(t) = \cos\left(\frac{t/T + s}{1 + s} \cdot \frac{\pi}{2}\right)^2$$

Variance learning (IDDPM, 2021)

$$p_{\theta}(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_{\theta}(x_t, t), \Sigma_{\theta}(x_t, t))$$

$$\text{with } \Sigma_{\theta}(x_t, t) = \sigma_t^2 I$$

$$\sigma_t^2 = \beta_t \text{ or } \sigma_t^2 = \tilde{\beta}_t$$

DDPM

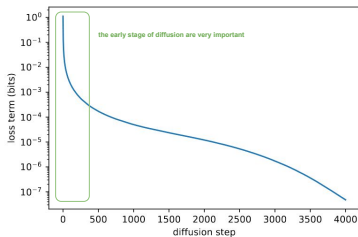


Limitations :

“(…) learning reverse process variances (…)

leads to unstable training and poorer sample quality compared to fixed variance.”

(DDPM, 2020)



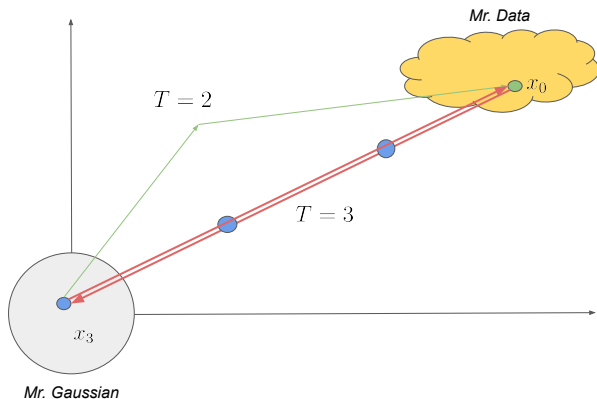
$$\Sigma_{\theta}(x_t, t) = \exp(v \log \beta_t + (1 - v) \log(\tilde{\beta}_t))$$

$$L_{\text{hybrid}} = L_{\text{simple}} + \lambda L_{\text{VLB}}$$

0.001

IDDPM

Diffusion & reverse process (DDIM, 2021)



Limitations :

“For example, it takes around 20 hours to sample 50k images of size 32 x 32 from a DDPM, but less than a minute to do so from a GAN on a Nvidia 2080 Ti GPU.” (DDIM, 2021)

Extension of DDPM (DDIM, 2021)

**Generalisation to a bigger class of
inverse process (non-Markovian)**

$$q_{\sigma}(x_{t-1}|x_t, x_0) = \mathcal{N}(\sqrt{\alpha_{t-1}}x_0 + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} \cdot \frac{x_t - \sqrt{\bar{\alpha}_t}x_0}{\sqrt{1 - \bar{\alpha}_t}}, \sigma_t^2 I)$$

$$\sigma_t^2 = \tilde{\beta}_t \quad \Rightarrow \quad DDPM$$

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I)$$



Important :

Same network and
training as a DDPM

Generation process (DDIM, 2021)

$$q_{\sigma}(x_{t-1}|x_t, x_0) = \mathcal{N}(\sqrt{\alpha_{t-1}}x_0 + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} \cdot \frac{x_t - \sqrt{\alpha_t}x_0}{\sqrt{1 - \alpha_t}}, \sigma_t^2 I)$$



DDIM



DDPM

$$x_{t-1} = \sqrt{\alpha_{t-1}}x_0 + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} \cdot \frac{x_t - \sqrt{\alpha_t}x_0}{\sqrt{1 - \alpha_t}} + \sigma_t z$$

$$x_{t-1} = \frac{\sqrt{\alpha_{t-1}}\beta_t}{1 - \bar{\alpha}_t}x_0 + \frac{\sqrt{\alpha_t}(1 - \alpha_{t-1})}{1 - \bar{\alpha}_t}x_t + \tilde{\beta}_t z$$

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon \implies x_{t-1} \text{ can be computed !!!}$$
$$\epsilon_{\theta}(x_t, t) \approx \epsilon$$

Finding a better inverse process (DDIM, 2021)

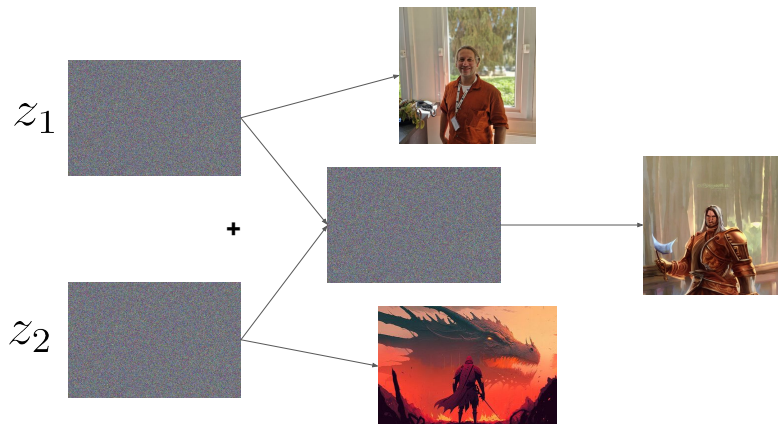
$$\sigma_t^2 = \eta_t \tilde{\beta}_t \quad \eta \in [0, 1]$$

$\eta = 0 \quad \Rightarrow \text{DDIM}$

S	CIFAR10 (32×32)					CelebA (64×64)					
	10	20	50	100	1000	10	20	50	100	1000	
η	0.0	13.36	6.84	4.67	4.16	4.04	17.33	13.73	9.17	6.53	3.51
	0.2	14.04	7.11	4.77	4.25	4.09	17.66	14.11	9.51	6.79	3.64
	0.5	16.66	8.35	5.25	4.46	4.29	19.86	16.06	11.01	8.09	4.28
	1.0	41.07	18.36	8.01	5.78	4.73	33.12	26.03	18.48	13.93	5.98
$\hat{\sigma}$	367.43	133.37	32.72	9.99	3.17	299.71	183.83	71.71	45.20	3.26	

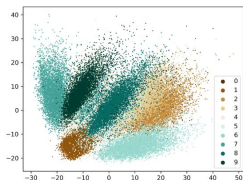
FID

Noise interpolation (DDIM, 2021)

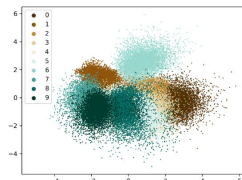


Latent diffusion : Concept of latent space (reminder)

Latent space of MNIST database for AE and VAE



(a) Latent Distribution by Label for AE



(b) Latent Distribution by Label for VAE

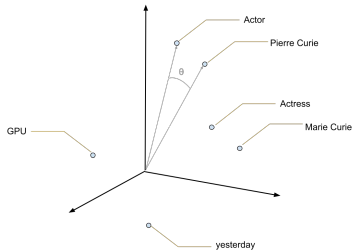
Source <https://thibaudsimon.com/visualizing-an-interpretable-latent-space/>

- Similar objects are close to one another in the latent space
- Usually lower dimension than original data (therefore does compression as well)
- Usually impossible to visualize by a human

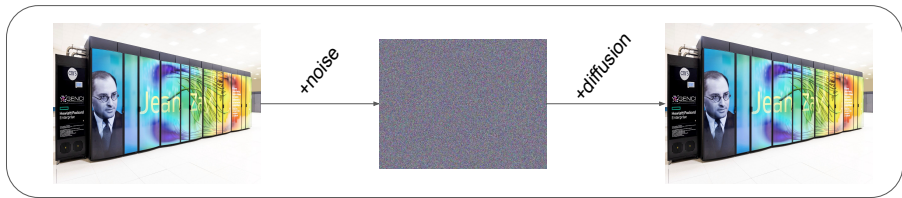
Latent diffusion : Concept of latent space (example)

Example : **Word embedding**

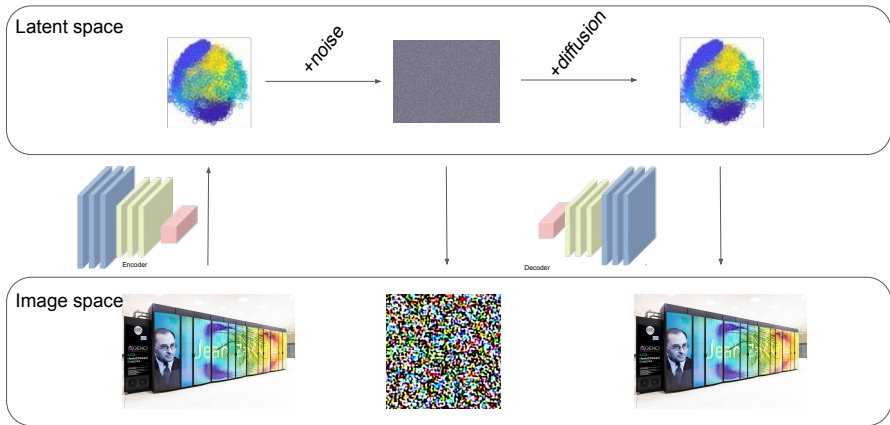
Turn sparse data (for instance words) into vectors



$$\text{Actor} - \text{Pierre Curie} + \text{Marie Curie} \approx \text{Actress}$$



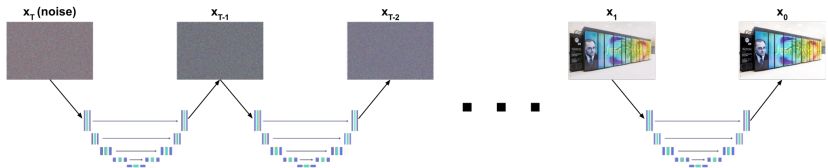
Latent diffusion model



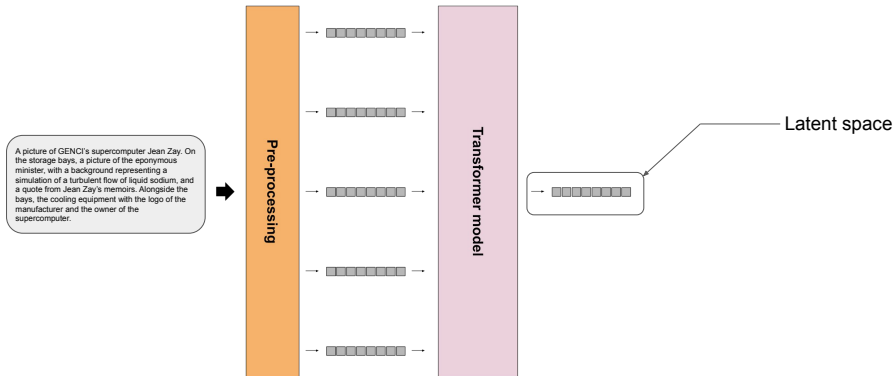
Conditional diffusion

How to control the output of a diffusion model and make sure it generates what we want ?

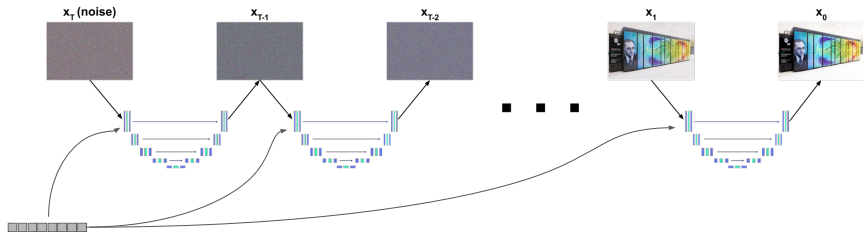
A picture of GENCI's supercomputer Jean Zay. On the storage bays, a picture of the eponymous minister, with a background representing a simulation of a turbulent flow of liquid sodium, and a quote from Jean Zay's memoirs. Alongside the bays, the cooling equipment with the logo of the manufacturer and the owner of the supercomputer.



Conditional diffusion : text $\rightarrow$ image

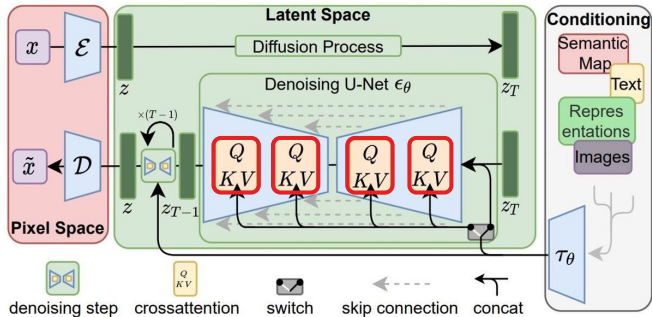


Conditional diffusion: text $\rightarrow$ image



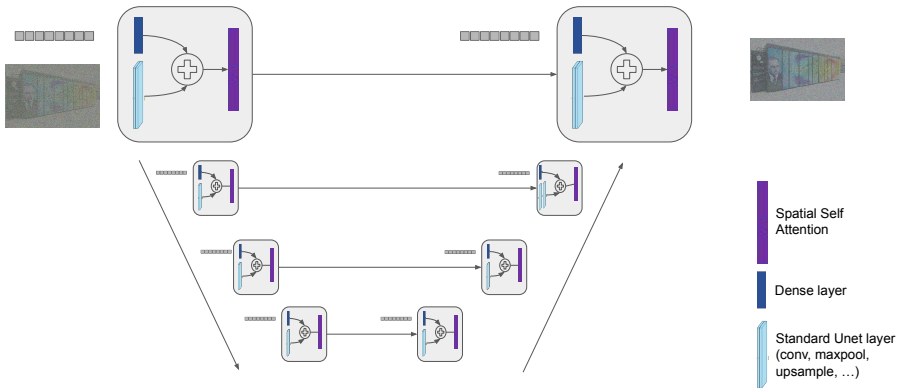
Conditional diffusion : cross-attention

Stable Diffusion uses cross-attention to make the denoising process consistent with the provided sentence embedding



Source Rombach, Robin, et al. "High-resolution image synthesis with latent diffusion models." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2022.

Conditional diffusion : other method



Other tasks

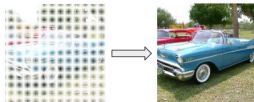
Diffusion models can solve a variety of tasks. We already know about image generation, as well as conditional image generation (for instance with a short paragraph describing the picture)

Other tasks:

→ Inpainting



→ Super-resolution



→ Outpainting



Inpainting through masking

We can solve many of these tasks through the usage of a mask

→ Wide mask



→ Thin mask



→ Outer mask for expanding the image



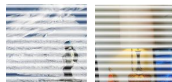
→ Right side mask for halving the image



→ Every second pixel in both directions for super-resolution

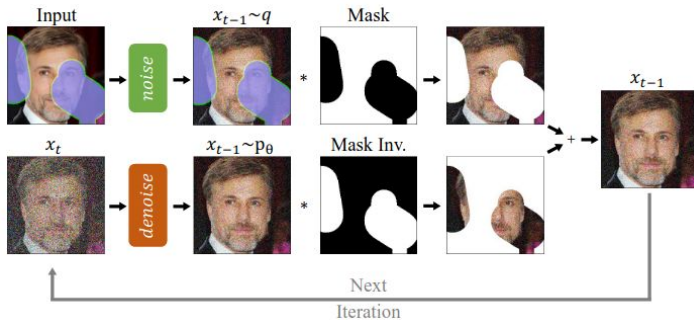


→ Every second row of pixels for alternating lines

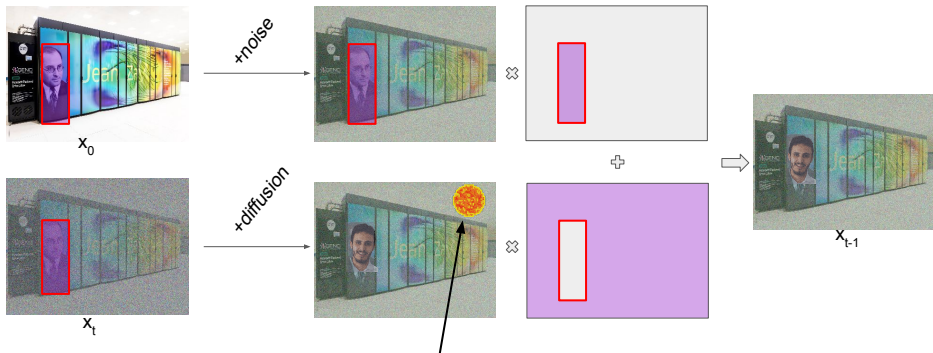


Inpainting through masking

Step t :

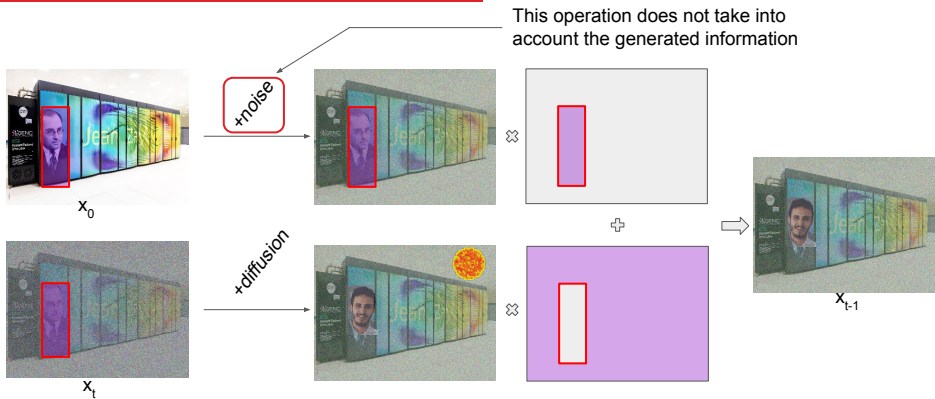


Inpainting through masking: step t



New artifacts added (in this coarse example, our diffusion model drew a sun), so we force the known background again!

Inpainting through masking: step t



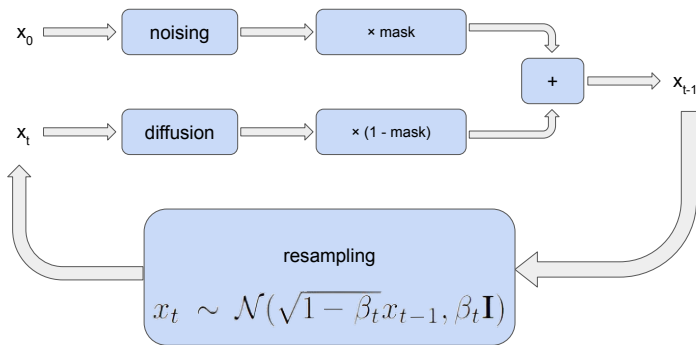
Inpainting deharmonization

Picture deharmonization: the generated image has a satisfying texture but is wrong semantically. The suggested solution is to resample.



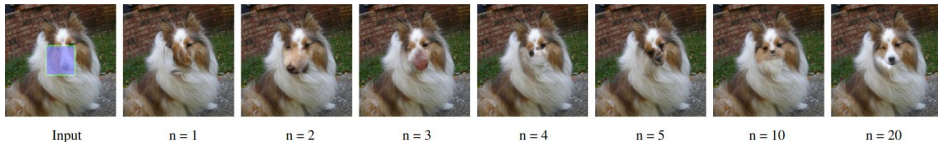
Source Lugmayr, Andreas, et al. "Repaint: Inpainting using denoising diffusion probabilistic models." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2022.

Inpainting resampling



This loop is performed several times (a hyperparameter) before moving on the next step

Inpainting resampling



n is the number of times the resampling loop was performed

Disadvantage: the number of required denoising steps is much higher

Advantage: it produces much more satisfying results

Sources

Papers:

- Deep Unsupervised Learning using Nonequilibrium Thermodynamics (DPM) (<https://arxiv.org/abs/1503.03585>)
- Denoising Diffusion Probabilistic Models (DDPM) (<https://arxiv.org/abs/2006.11239>)
- Improved Denoising Diffusion Probabilistic Models (IDDP) (<https://arxiv.org/abs/2102.09672>)
- Denoising Diffusion Implicit Models (DDIM) (<https://arxiv.org/abs/2010.02502>)
- Diffusion Models Beat GANs on Image Synthesis (<https://arxiv.org/abs/2105.05233>)
- High-Resolution Image Synthesis with Latent Diffusion Models (LDM) (<https://arxiv.org/abs/2112.10752>)
- Repaint: Inpainting using denoising diffusion probabilistic models (<https://arxiv.org/pdf/2201.09865>)
- Diffusion Models in Vision: A Survey (<https://arxiv.org/abs/2209.04747>)
- Diffusion Models: A Comprehensive Survey of Methods and Applications(<https://arxiv.org/abs/2209.00796>)

Other ressources:

- Lilian Weng's article (<https://lilianweng.github.io/posts/2021-07-11-diffusion-models>)
- Yang Song's article (<https://yang-song.net/blog/2021/score>)
- Outlier video (<https://www.youtube.com/watch?v=HoKDTa5jHvg>)

Diffusion process and Flow Matching - Overview (1/3)

- ▶ A Diffusion process destroys an observed data point x_0 over time t by introducing Gaussian noise
 - ▶ $z_t = \alpha_t x_0 + \sigma_t \epsilon$ with $\epsilon \sim \mathcal{N}(0, 1)$, α_t and σ_t are noise schedule, $t \in [0, 1]$ for convenience, z_0 close to clean data, z_1 close to (pure) Gaussian noise.
 - ▶ To sample a new instance, we initialize z_1 with standard Gaussian, given a sample at time step t , we predict (alternatively) the clean sample or the noise level $\hat{\epsilon}$.
 $z_s = \alpha_s(\hat{x}_0) + \sigma_s \hat{\epsilon}$, $s < t$, $\hat{x}_0 = \frac{1}{\alpha_t}(z_t - \sigma_t \hat{\epsilon})$.
- ▶ Can be described by a Stochastic Differential Equation (SDE):

$$dz_t = f_t z_t dt + g_t dz$$

dz is a Brownian motion, f_t and g_t are noise schedule.

- ▶ The generative process is given by the reverse of the forward process as: $dz_t = \left(f_t z_t - \frac{1+\eta_t^2}{2} g_t^2 \nabla \log p_t(z_t) \right) dt + \eta_t g_t dz$
 η_t : parameter for the stochasticity at inference time
- ▶ $\nabla \log p_t$ is called the *score* of the forward process
- ▶ Other names: Score matching or Score-based Generative modeling

Diffusion process and Flow Matching - Overview (2/3)

- ▶ Flow matching: linear interpolation between data and noise term
 - ▶ $z_t = (1 - t)x_0 + t\epsilon$
 - ▶ Corresponds to diffusion process if the noise is Gaussian and we use schedule $\alpha_t = 1 - t, \sigma_t = t$
 - ▶ We can derive $z_t = z_s + u * (t - s)$, for $s < t$, where $u = \epsilon - x$.
 - ▶ u is the velocity field that is learned by a neural network
$$\hat{u} = \hat{u}(z_t; t) = \hat{\epsilon} - \hat{x}$$
$$z_s = z_t + \hat{u} * (s - t)$$
 - ▶ The interpolation between x and ϵ can be described by an Ordinary Differential Equation (ODE):
$$\frac{dz_t}{dt} = u_t$$
Assuming $z_t = \alpha_t x + \sigma_t \epsilon$ then $u_t = \dot{\alpha}_t x + \dot{\sigma}_t \epsilon$, where $\dot{x}$ denotes the derivative of x wrt time (t)

Diffusion process and Flow Matching - Overview (3/3)

- ▶ The two principles are equivalent. (Gaussian) Flow Matching brings
 - ▶ a vector field parameterization of the network output. Can be interesting when higher order samplers are used.
 - ▶ Simple sampling noise schedules

Conditional flow matching [Lipman, 2022] (without justification)

Let u_θ a neural network

Repeat

1. Draw $x_0 \sim N(0, I)$
2. Draw $x_1 \sim p_{data}$
3. Sample $t \in [0, 1]$
4. Let $x_t \leftarrow (1 - t)x_0 + tx_1 + N(0, I)$
5. minimize $_\theta \|u_\theta(x_t) - (x_0 - x_1)\|^2$

To generate an image, solve the ODE $\frac{\partial \phi(t)}{\partial t} = u_\theta(x_t)$

Ethics and Fairness in Machine Learning

We some materials coming from the Fidle course on Ethics in AI (15)
<https://fidle.cnrs.fr>, previous Materials from Charlotte Laclau

A story - from Jessica Hutt

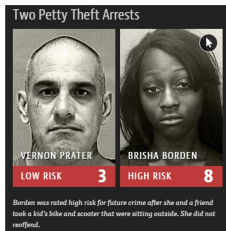
- ▶ An engineering professor gave students an assignment to design a pipeline to send blood from a poor developing country to a rich developed one
- ▶ The students got to work, discussing the optimal diameters for the pipe, how to go under a body of water, etc.
- ▶ After letting them discuss for a while, the professor asked to know why not one of them had questioned the ethics of the assigned task
- ▶ *"This is a class in engineering not ethics"* was the answer given by the students.

⇒ Ask **WHY** before **HOW**
Questions?

- ▶ Is the task ethical? What the pros and cons?
- ▶ Who might benefit? Who might suffer?

Discrimination issues

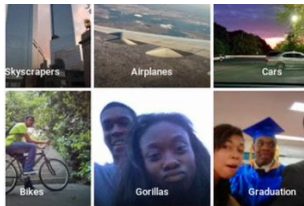
COMPAS



Source : *ProPublica*

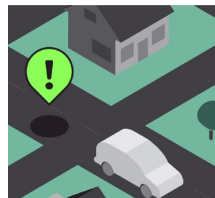


GOOGLE PHOTOS



Source : *The Verge*

STREET BUMP



Source : *Boston.gov*

Ronan Pons (CNRS, 3IA ANITI)

From the Fidle course on Ethics and Law <https://fidle.cnrs.fr>

Ethics issues

- ▶ AI hiring tools show bias against women
- ▶ Algorithms used by judges to decide who is likely to commit (another) crime
- ▶ Medical diagnosis algorithms trained on data excluding certain sub-populations.
- ▶ Algorithms to identify gay people.
- ▶ Automatic evaluation of citizens in China.

9 Principles for Ethics AI

- ▶ International Human Rights
- ▶ Promotion of Human Values
- ▶ Professional Responsibility
- ▶ Human Control of Technology
- ▶ Fairness and Non-discrimination
- ▶ Transparency and Explainability
- ▶ Safety and Security
- ▶ Accountability
- ▶ Privacy

Discrimination issues

PRINCIPLED ARTIFICIAL INTELLIGENCE

A Map of Ethical and Rights-Based Approaches to Principles for AI

Authors: Jessica Fjeld, Nele Achten, Hannah Hilligoss, Adam Nagy, Madhulika Srikanth

Designers: Arushi Singh (arushisingh.net) and Melissa Axelrod (melissaaxelrod.com)

HOW TO READ:

Date, Location
Document Title
Author

COVERAGE OF THEMES:



The size of each dot represents the percentage of principles in that theme contained in the document. Since the number of principles per theme varies, it's informative to compare dot sizes within a theme but not between themes.

The principles within each theme are:

[Privacy](#)

- Privacy
- Control over Use of Data
- Consent
- Privacy by Design
- Recommendation for Data Protection Laws
- Ability to Restrict Processing
- Right to Rectification
- Right to Erasure

Accountability:

- Accountability
- Recommendation for New Regulations
- Impact Assessment
- Evaluation and Auditing Requirement
- Verifiability and Replicability
- Liability and Legal Responsibility
- Ability to Appeal
- Environmental Responsibility
- Creation of a Monitoring Body
- Remedy for Automated Decision

Safety and Security

- Security
- Safety and Reliability
- Predictability
- Security by Design

Transparency and Explainability:

- Explainability
- Transparency
- Open Source Data and Algorithms
- Notification when Interacting with an AI
- Notification when AI Makes a Decision about an Individual
- Regular Reporting Requirement
- Right to Information
- Open Procurement (for Government)

Fairness and Non-discrimination:
Non-discrimination and the Prevention of Bias
Fairness
Inclusiveness in Design
Inclusiveness in Impact
Representative and High Quality Data
Equality

Human Control of Technology:
Human Control of Technology
Human Review of Automated Decision
Ability to Opt out of Automated Decisions

Professional Responsibility:
Multistakeholder Collaboration
Responsible Design
Consideration of Long-Term Effects

Accuracy
Scientific integrity

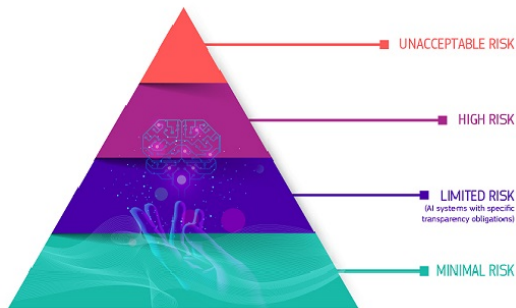
Promotion of Human Values:
Leveraged to Benefit Society
Human Values and Human Flourishing
Access to Technology

➤ | ➤ | ➤

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The 4 main categories of risks in AI



- ▶ Unacceptable risk
- ▶ High risk
- ▶ Limited risk
- ▶ Minimal or no risk

Unacceptable risk

All AI systems considered a clear threat to the safety, livelihoods and rights of people will be banned, from social scoring by governments to toys using voice assistance that encourages dangerous behaviour.

High risk

AI systems identified as high-risk include AI technology used in:

- ▶ critical infrastructures (e.g. transport), that could put the life and health of citizens at risk;
- ▶ educational or vocational training, that may determine the access to education and professional course of someone's life (e.g. scoring of exams);
- ▶ safety components of products (e.g. AI application in robot-assisted surgery);
- ▶ employment, management of workers and access to self-employment (e.g. CV-sorting software for recruitment procedures);
- ▶ essential private and public services (e.g. credit scoring denying citizens opportunity to obtain a loan);
- ▶ law enforcement that may interfere with people's fundamental rights (e.g. evaluation of the reliability of evidence);
- ▶ migration, asylum and border control management (e.g. verification of authenticity of travel documents);
- ▶ administration of justice and democratic processes (e.g. applying the law to a concrete set of facts).

High risk

High-risk AI systems will be subject to strict obligations before they can be put on the market:

- ▶ adequate risk assessment and mitigation systems;
- ▶ high quality of the datasets feeding the system to minimise risks and discriminatory outcomes;
- ▶ logging of activity to ensure traceability of results;
- ▶ detailed documentation providing all information necessary on the system and its purpose for authorities to assess its compliance;
- ▶ clear and adequate information to the user;
- ▶ appropriate human oversight measures to minimise risk;
- ▶ high level of robustness, security and accuracy.

High risk

All remote biometric identification systems are considered high risk and subject to strict requirements. The use of remote biometric identification in publicly accessible spaces for law enforcement purposes is, in principle, prohibited.

Narrow exceptions are strictly defined and regulated, such as when necessary to search for a missing child, to prevent a specific and imminent terrorist threat or to detect, locate, identify or prosecute a perpetrator or suspect of a serious criminal offence.

Such use is subject to authorisation by a judicial or other independent body and to appropriate limits in time, geographic reach and the data bases searched.

Limited risk

Limited risk refers to AI systems with specific transparency obligations. When using AI systems such as chatbots, users should be aware that they are interacting with a machine so they can take an informed decision to continue or step back.

Minimal or no risk

The proposal allows the free use of minimal-risk AI. This includes applications such as AI-enabled video games or spam filters. The vast majority of AI systems currently used in the EU fall into this category.

System Monitoring



Once an AI system is on the market, authorities are in charge of market surveillance, users ensure human oversight and monitoring, and providers have a post-market monitoring system in place. Providers and users will also report serious incidents and malfunctioning.

Fairness in Machine Learning

We use the slides of C. Laclau

WHAT DOES THE LAW SAYS ?

HOW TO DEFINE FAIRNESS ?

- ▶ What does the law say ? In France

Article 225-1 of the penal code

Discrimination is any distinction made between persons on the basis of their origin, sex, marital status, pregnancy, physical appearance, economic status, place of residence, surname, health status, loss of autonomy, disability, genetic characteristics, morals, sexual orientation, gender identity, age, political opinion, their ability to express themselves in a language other than french, their membership or non-membership, true or supposed, to an ethnic group, a nation, an alleged race or a specific religion.

HOW TO DEFINE FAIRNESS ?

► What does the law say ? In France

Article 225-2 of the penal code

The discrimination defined in Article 225 committed against a person is punishable by 3 years of imprisonment and a fine of 45 000 euros when it consists of:

- refusing the supply of a good or service
- hindering the normal exercise of any economic activity
- refusing to hire, sanctioning or firing a person

HOW TO DEFINE FAIRNESS ?

► What does the law say ? In the EU

Article 22§4

The discrimination defined in Article 225 committed against a person is punishable by 3 years of imprisonment and a fine of 45 000 euros when it consists of:

- refusing the supply of a good or service
- hindering the normal exercise of any economic activity
- refusing to hire, sanctioning or firing a person

HOW TO DEFINE FAIRNESS ?

► What does the law say ? In the US (definitions)

Regulated Domains include Lending, Education, Hiring, Housing (extends to target advertising).

Protected (Sensitive) Groups : Reflect the fact that in the past there have been unjust practices.

Disparate Treatment

Individuals are treated differently because of protected characteristics (e.g. race or gender).

[Equal Protection Clause of the 14th Amendment.]

Disparate Impact

An apparently neutral policy that adversely affects a protected group more than another group.

[Civil Rights Act, Fair Housing Act, and various state statutes.]

CAN WE COME UP WITH A MATHEMATICAL DEFINITION OF FAIRNESS ?

RECALL - SUPERVISED SETTING

Let S be a set of m **i.i.d.** training examples $\{z_i = (\mathbf{x}_i, y_i)\}_{i=1}^m$ from an **unknown** joint distribution $D_{\mathcal{Z}}$ over $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$.

1. The $\mathbf{x}_i$ values ($\mathbf{x}_i \in \mathcal{X}$) are typically vectors of the form $(x_{i1}, \dots, x_{id})$, whose components are usually called **features**.
2. The y values ($y \in \mathcal{Y}$) are drawn from a **discrete set** of **classes** (typically $\mathcal{Y} = \{-1, +1\}$ in binary classification) or are continuous values (regression).
3. We assume that there exists a **target function** f such that for $(\mathbf{x}, y) \in \mathcal{Z}$

$$y = f(\mathbf{x})$$

Definition

A supervised learning algorithm outputs from S a classifier (or hypothesis) $h \in \mathcal{H}$ about the target function f .

RECALL - CLASSICAL METRIC FOR CLASSIFICATION

Traditional approach
is to maximize

$$\text{Precision} = TP / (TP + FP)$$

$$\text{Recall} = TP / (TP + FN)$$

$$\text{Specificity} = TN / (FP + TN)$$

Accuracy =

$$(TP + TN) / (TP + TN + FP + FN)$$

		Actual Condition		
		Total Samples		
Output of Classifier	Classify Positive	TP	FP	PPV (precision)
	Classify Negative	FN	TN	
		TPR (Recall)	TNR (Specificity)	ACC
				F-measure
				MCC

WHY IS IT NOT ENOUGH TO HAVE HIGH ACCURACY ?

- ▶ Let us consider a classifier that is used by an organisation to hire 100 candidates who applied (69 male and 31 females)
- ▶ If the classifier predicts 1 ($Y'=1$) then organization will hire the candidate
- ▶ **True class** $Y=1$ if the candidate really deserves to be hired, 0 otherwise
- ▶ Classifier achieves an **overall accuracy of 90%**

Overall	$Y = 1$	$Y = 0$
$Y' = 1$	46	4
$Y' = 0$	6	44

- ▶ For males, the accuracy rises to 94.2%

Males	$Y = 1$	$Y = 0$
$Y' = 1$	35	4
$Y' = 0$	0	30

- ▶ For females, the accuracy drops to 80.64%

Females	$Y = 1$	$Y = 0$
$Y' = 1$	11	0
$Y' = 0$	6	14

- ▶ What is the problem here ?

Males	Y = 1	Y = 0
Y' = 1	35	4
Y' = 0	0	30

Females	Y = 1	Y = 0
Y' = 1	11	0
Y' = 0	6	14

4 Undeserving male
candidates got
hired!!!

6 Deserving female
candidates got
rejected!!!

- ▶ Misclassification for a system of such large consequence costs a lot
- ▶ From the organisation's point of view, they hired 4 undeserving candidates
- ▶ From the candidates' perspective, 6 deserving candidates got rejected

- ▶ Also, is this system **legal** (in the US) ?
 - ▶ **NO** (According to U.S. Equal Employment Opportunity Commission: the "80%-rule")
 - ▶ Selection rate for males = 56.52% (39 out of 69)
 - ▶ Selection rate for females = 35.48% (11 out of 31)
 - ▶ **According to the rule, the selection** rate for females should be at least 80% of the selection rate for males

We need some way to measure/define fairness (just like we measure performance)

BIAS, DISCRIMINATION, FAIRNESS

- ▶ **Bias:** Inclination / prejudice for or against one person or group, especially in a way considered to be unfair.
- ▶ **Discrimination:** The unjust or prejudicial treatment of different categories of people, especially on the grounds of race, age, or sex.
- ▶ **Fairness:** Absence of any prejudice or favoritism toward an individual or a group based on their inherent or acquired characteristics

Let's focus on the fairness term in the classification context

MULTIPLE TYPE OF FAIRNESS

- ▶ **Individual Fairness** : different individuals who are (almost) equal in various aspects should be treated (almost) equally
 - ▶ E.g., if candidates A and B have similar qualifications, it should not be that A is selected but B is rejected
- ▶ **Group Fairness**: different groups should not be treated too differently
 - ▶ E.g., selection rate of females should not be too much lower than that for males

Definition of fairness varies according to type of fairness, the domain and the law

COMMON METRICS

- ▶ Let be p the protected attribute (e.g. the gender) s.t. $p_i = 0$ if the x_i - th observation is a male and 1 a female
- ▶ Fairness can be expressed as a function of

- ▶ The target function f
 - ▶ The protected attribute p
 - ▶ The training labels y_i
-
- | | | | |
|--|--|--|----------------------|
| | | | Individual fairness |
| | | | Demographic parity |
| | | | Equalized error rate |

COMMON METRICS

- ▶ **Individual Fairness** : $D(f(x), f(x')) \leq d(x, x')$
- ▶ **Demographic parity** : $Pr[f(x) = 1 | p = 1] \approx Pr[f(x) = 1 | p = 0]$
- ▶ **Equalized error rates**: $Pr[f(x) \neq y | p = 1] \approx Pr[f(x) \neq 1 | p = 0]$
- ▶ **Group sensitive calibration**: $\forall g, r, Pr[y = 1 | f(x) = r, p = g] = r$

NORMATIVE DIMENSION TO FAIRNESS

Normative: establishing, standardizing or pertaining to a norm

All measures of fairness carry normative positioning within them

- ▶ **Individual fairness:** there exists an objective measure of ability and fairness is making sure people of similar ability are treated similarly
- ▶ **Demographic parity:** group identity should have nothing to do with selection for a task.
- ▶ **Equalized odds:** groups may have different innate skill levels, but we should make mistakes equally.

EXTENSION TO REGRESSION

Given $(x_i, y_i), x_i \in \mathbb{R}^d, y_i \in \mathbb{R}$, we aim at finding a mapping $f: \mathbb{R}^d \rightarrow \mathbb{R}$ such that

$$\min_f \sum_i ||f(x_i), y_i||^2$$

- ▶ What is an appropriate form of fairness ?
- ▶ What is the context for this ? (eg. Credit score assignment)
- ▶ What would the right normative concerns be ?

Still an open challenge !

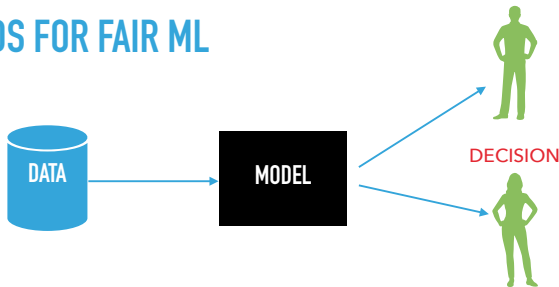
EXTENSION TO RANKING

We are interested in building a ranking scheme to rank individuals from top to bottom (for example for hiring)

- ▶ What are forms of fairness in rankings ?
 - ▶ "demographic parity": proportion of people from different groups in each top-k section should be roughly the same
 - ▶ "quotas": each group should have an upper/lower bound on number of members in top-k for different k
 - ▶ generalize rank to "exposure" and equalize it

- ▶ About the lecture
- ▶ Motivation
- ▶ Machine Learning setting
- ▶ Definitions of Fairness
- ▶ **How to satisfy fairness in ML ?**

METHODS FOR FAIR ML



- ▶ **Pre-processing** : modify the training data or the representation of the data to remove the bias prior to the training process
- ▶ **In-processing** : Enforce fairness notions by imposing constraints into the optimization, or by using an adversary.
- ▶ **Post-processing**: post-process the model outputs

▶ **Pre-processing**

- ▶ PRO: Might be able to account for data bias, deal with black box classifier
- ▶ CON: Might need to modify data extensively to remove skew - this is not well defined.
- ▶ CON: no guarantee that sophisticated models will not recapture the bias

▶ **In-processing**

- ▶ PRO: Flexibility to work with any data. More control over the learned model
- ▶ CON: all regularizers are proxies / optimization issues which make also the outcome explanation more difficult

▶ **Post-processing**

- ▶ PRO: Works with black box classifier and any training data. Has certain optimality properties
- ▶ CON: still illegal in the US in certain sectors

Conclusion

- ▶ Importance of latent representations to use/combine Deep Learning models
- ▶ Can handle a large spectrum of objects
- ▶ Need for less costly/frugal methods
- ▶ Are in continuous development!
- ▶ Importance of Ethics and Fairness issues.